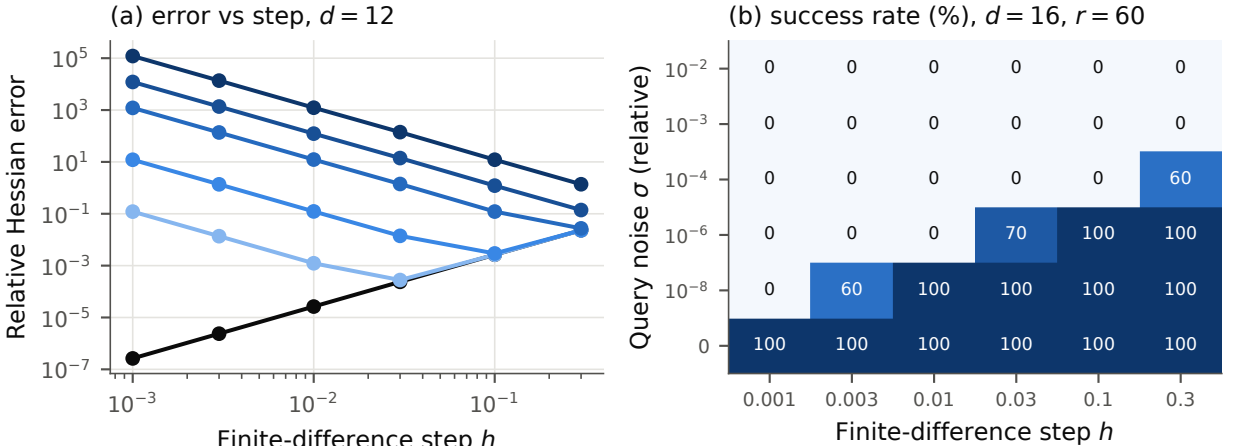
(a) error vs step, $d = 12$
Relative Hessian error
10^{5}
10^{3}
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-3}
10^{-2}
10^{-1}
Finite-difference step h
$\sigma = 0$
$\sigma = 10^{-8}$
$\sigma = 10^{-6}$
$\sigma = 10^{-4}$
$\sigma = 10^{-3}$
$\sigma = 10^{-2}$
(b) success rate (%), $d = 16$, $r = 60$
Query noise σ (relative)
10^{-2}
10^{-3}
10^{-4}
10^{-6}
10^{-8}
0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 60
0 0 0 70 100 100
0 60 100 100 100 100
100 100 100 100 100 100
0.001
0.003
0.01
0.03
0.1
0.3
Finite-difference step h